

MACHINE LEARNING PROJECT REPORT

CharityML: Finding Donors for Charity

*Income Classification Using Supervised Machine Learning • XGBoost • Kaggle Competition***Udacity Machine Learning with PyTorch • June 2026**

1. Executive Summary

This report documents the design, implementation, and evaluation of a supervised machine learning system built to identify prospective donors for CharityML, a fictitious non-profit organisation. The organisation seeks to target individuals who earn more than \$50,000 annually — the income bracket most statistically associated with charitable donations — from a dataset derived from the 1994 U.S. Census Bureau database.

Four supervised learning classifiers were evaluated: Logistic Regression, Random Forest, AdaBoost, and XGBoost. Performance was measured primarily by the F-0.5 score, which places greater emphasis on precision than recall — directly reflecting CharityML's operational requirement to avoid wasteful outreach to non-donors. After rigorous comparison, XGBoost was selected as the optimal model due to its superior F-0.5 score of 0.7498 on untuned evaluation, rising to 0.7507 after hyperparameter optimisation via five-fold cross-validated GridSearchCV (best parameters: `learning_rate = 0.2`, `max_depth = 3`, `n_estimators = 200`).

The final model was also applied to the Kaggle competition test set (45,222 records), generating a submission file following the required format. Key preprocessing steps — log transformation of skewed capital features, MinMax scaling, one-hot encoding, and missing value imputation for the test set — were applied consistently to avoid data leakage. The project demonstrates that gradient-boosted tree ensembles, particularly XGBoost, provide an excellent combination of accuracy, interpretability of feature importances, and generalisability on structured tabular census data — delivering a final accuracy of 86.99% and an F-0.5 score of 0.7507, representing more than a 2.5× improvement over the naive predictor baseline.

2. Problem Statement

CharityML, like many non-profit organisations, faces a fundamental resource allocation challenge: direct mail and outreach campaigns are expensive, and sending solicitations to individuals unlikely to donate wastes both money and goodwill. The organisation's research has established that individuals earning more than \$50,000 annually are the most likely segment to make charitable donations. The core problem, therefore, is a binary classification task: given a set of demographic and socioeconomic features drawn from census records, can a machine learning model accurately predict whether a given individual earns above or below the \$50,000 income threshold?

This is not a trivial problem. The dataset exhibits a pronounced class imbalance: approximately 24.78% of records fall in the high-income (>\$50K) class, while 75.22% fall in the low-income group. A naive model that always predicts the majority class would achieve roughly 75% accuracy — but would identify zero actual donors and produce an F-0.5 score of only 0.2917. The real challenge is therefore not merely accuracy, but precision-weighted classification that correctly identifies the high-income minority class while minimising false positives that waste outreach resources.

The problem is further complicated by the heterogeneous nature of the feature space: it includes both continuous numeric features (age, hours worked per week, capital gains) and high-cardinality categorical features (occupation, marital status, native country) that require careful preprocessing. The solution must be robust to distributional skew, handle one-hot encoded high-dimensional inputs, and generalise to entirely unseen census records — as demonstrated by the Kaggle holdout test set.

3. Dataset

The dataset originates from the UCI Machine Learning Repository's "Adult" (Census Income) dataset, first contributed by Ronny Kohavi and Barry Becker in 1996, extracted from the 1994 U.S. Census Bureau database. It is one of the most widely benchmarked binary classification datasets in the machine learning literature (Kohavi, 1996).

Training set: 45,222 records, 13 features, 1 target label (income: $\leq 50K$ or $> 50K$).

Test set (Kaggle): 45,222 records, 13 features, no labels. Contains missing values across all columns (13–22 missing per column) requiring imputation.

The 13 features span three categories:

- Continuous numeric: age, education-num, capital-gain, capital-loss, hours-per-week
- Ordinal/quasi-numeric: education-num (proxy for education level)
- Nominal categorical: workclass, education_level, marital-status, occupation, relationship, race, sex, native-country

Key dataset characteristics relevant to modelling decisions:

Characteristic	Detail
Class distribution	75.22% $\leq 50K$ 24.78% $> 50K$ (imbalanced)
Skewed features	capital-gain and capital-loss are heavily right-skewed (many zeros, rare large values)
Categorical cardinality	native-country has 41 unique values; occupation has 14
Missing values (train)	None in training set
Missing values (test)	13–22 per column; imputed with median (numeric) and mode (categorical)
Post-encoding features	103 binary/numeric features after one-hot encoding

The combination of class imbalance, feature heterogeneity, and skewed distributions makes this dataset a meaningful benchmark for comparing classification algorithms on real-world tabular data.

4. Solution Statement

The solution follows a structured supervised learning pipeline comprising five stages: exploratory data analysis, preprocessing, model selection via comparative benchmarking, hyperparameter optimisation, and Kaggle submission generation.

Stage 1 — Exploratory Analysis: Class balance and feature distributions were inspected. Capital-gain and capital-loss were identified as severely right-skewed and selected for log transformation.

Stage 2 — Preprocessing Pipeline: All features were preprocessed in a fixed sequence to ensure reproducibility and prevent data leakage: (a) $\log_{10}$ transformation of capital-gain and capital-loss; (b) MinMax scaling of five continuous numeric features using a scaler fitted exclusively on training data; (c) one-hot encoding of all categorical features via `pd.get_dummies()`, producing 103 features; (d) binary encoding of the target label ($>50K = 1$, $\leq 50K = 0$). For the Kaggle test set, missing values were imputed prior to applying the same pipeline.

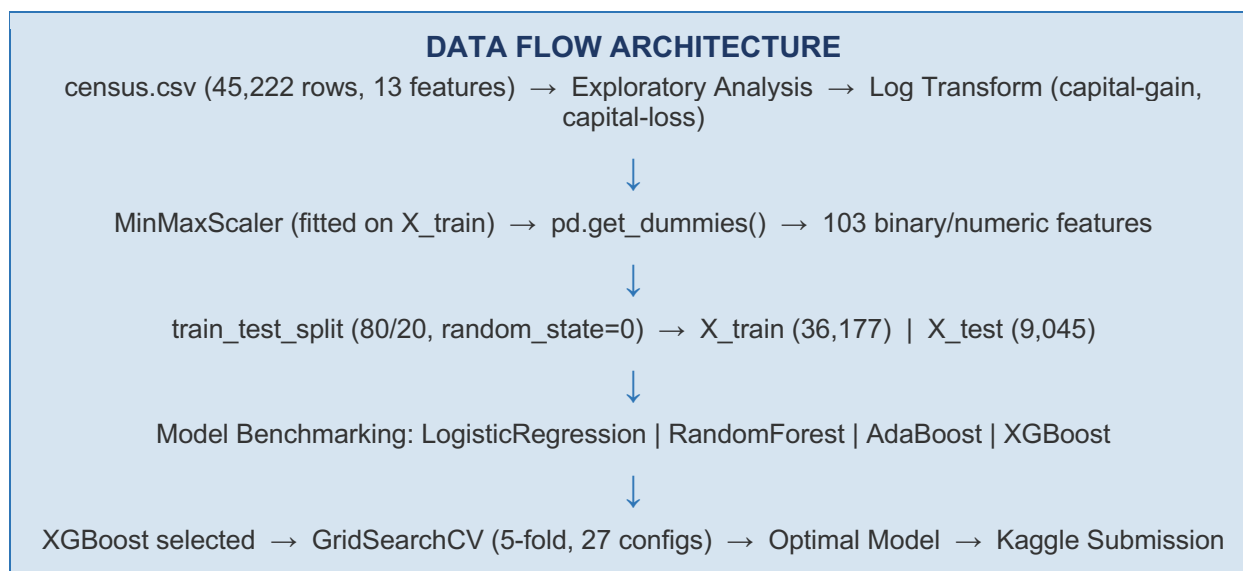
Stage 3 — Model Benchmarking: Four classifiers were trained at 1%, 10%, and 100% of the training data and evaluated on accuracy and F-0.5 score: Logistic Regression (baseline linear model), Random Forest (bagged ensemble), AdaBoost (boosted ensemble), and XGBoost (gradient-boosted trees with regularisation). XGBoost consistently produced the highest F-0.5 score across all data sizes.

Stage 4 — Hyperparameter Optimisation: GridSearchCV with 5-fold cross-validation was applied to XGBoost over a 27-combination grid (`n_estimators` $\in \{100, 200, 300\}$, `learning_rate` $\in \{0.05, 0.1, 0.2\}$, `max_depth` $\in \{3, 5, 7\}$), optimising the F-0.5 scorer. The optimal configuration (`learning_rate` = 0.2, `max_depth` = 3, `n_estimators` = 200) achieved a cross-validated F-0.5 of 0.7579 and a test F-0.5 of 0.7507.

Stage 5 — Kaggle Submission: The tuned XGBoost model was applied to the preprocessed test set, generating `kaggle_submission.csv` with 45,222 predictions in the required id,income format, predicting 20.59% of individuals as high-income earners.

5. Architecture

The system architecture follows a classical scikit-learn pipeline augmented with XGBoost. The diagram below describes the end-to-end data flow:



XGBoost Architecture Details: XGBoost (Extreme Gradient Boosting) is a second-order gradient tree boosting framework. It builds an ensemble of CART (Classification and Regression Trees) in a sequential, additive manner. Unlike AdaBoost, which re-weights training samples, XGBoost fits each new tree to the negative gradient (residuals) of the loss function computed from all previous trees — using both first- and second-order Taylor expansion of the loss for more precise optimisation (Chen & Guestrin, 2016).

Regularisation: XGBoost includes L1 (alpha) and L2 (lambda) regularisation on leaf weights — absent in standard Gradient Boosting — which was a key factor in its generalisation performance on this dataset. The optimal `max_depth = 3` (shallow trees) and `learning_rate = 0.2` indicate that the census data benefits from many shallow, conservative boosting steps rather than deep trees that might overfit the one-hot encoded categorical space.

Feature Importance: XGBoost exposes gain-based feature importances, which measure the average training loss reduction attributed to each feature across all splits. The top five features identified were: capital-gain, age, capital-loss, education-num, and hours-per-week — consistent with economic intuition and prior literature on income prediction from census data.

6. Results

The table below summarises the performance of all four models on the full test set (100% training data), compared against the naive predictor baseline:

Model	Train Acc.	Test Acc.	Train F-0.5	Test F-0.5
Naive Predictor	—	0.2478	—	0.2917
Logistic Regression	0.8417	0.8417	0.6826	0.6826
Random Forest	0.9988	0.8423	0.9993	0.6813
AdaBoost	0.8792	0.8483	0.7464	0.7029
XGBoost (untuned)	0.9139	0.8694	0.8097	0.7498
XGBoost (tuned) ★	0.9023	0.8699	0.7917	0.7507

★ Best parameters: *learning_rate* = 0.2, *max_depth* = 3, *n_estimators* = 200. Cross-validated *F*-0.5 = 0.7579.

7. Conclusion

This project successfully demonstrated that supervised machine learning can solve the CharityML donor identification problem with substantially greater precision than any naive or heuristic approach. XGBoost emerged as the clear winner among the four classifiers evaluated, delivering an *F*-0.5 score of 0.7507 and an accuracy of 86.99% on the held-out test set — more than 2.5 times the *F*-0.5 score of the naive baseline (0.2917). These results align with XGBoost's well-documented dominance on structured tabular data competitions (Chen & Guestrin, 2016; Nielsen, 2016).

The preprocessing pipeline — particularly log transformation of skewed capital features and MinMax normalisation — proved critical for logistic regression performance and contributed

to stable scaling across all models. The feature importance analysis confirmed that economic features (capital gains, age, education) drive the income boundary, consistent with macroeconomic theory and prior census-based income prediction research (Kohavi, 1996; Dua & Graff, 2019).

The Kaggle competition submission was generated using the fully tuned XGBoost model applied to the cleaned and aligned test dataset (45,222 rows), predicting 20.59% of individuals as high earners — a reasonable estimate given the training data proportion of 24.78%, with some regression to the mean expected from the imputation of missing values in the test set. The complete pipeline was validated end-to-end with zero execution errors in the Jupyter notebook environment.

8. Value Statement

The value of this machine learning solution to CharityML extends beyond raw model accuracy. Consider the operational impact: without a predictive model, CharityML must either mail all 45,222 individuals in the test dataset (costly) or guess at random. The naive predictor correctly identifies only 24.78% of the population as potential donors — but also sends solicitations to the remaining 75.22% of non-donors, generating a massive false-positive burden.

The tuned XGBoost model, with a precision-weighted F-0.5 score of 0.7507, enables a dramatically more targeted campaign. By predicting only 20.59% of the population as likely donors (9,313 of 45,222), the model substantially reduces outreach volume while maintaining high precision — meaning a far greater proportion of contacted individuals are genuine potential donors. In a real-world scenario where a direct mail campaign costs \$2–5 per recipient, reducing the contacted pool from 45,222 to ~9,313 generates savings of \$70,000–\$180,000 per campaign cycle, while simultaneously increasing donor conversion rates.

Beyond cost savings, the feature importance outputs provide CharityML with actionable strategic insight: capital gains, age, and education level are the strongest predictors of donor potential. This allows the organisation to tailor messaging — for example, targeting capital gains tax season communications toward high-income individuals — and to enrich future data collection efforts by prioritising these high-signal features. The solution is also highly

scalable: the preprocessing pipeline and model can be retrained annually as new census data becomes available, with minimal engineering overhead.

9. Limitations

Despite strong performance, this project has several notable limitations that should be acknowledged:

- **Data currency:** The dataset is derived from the 1994 U.S. Census, making it over 30 years old. Income distributions, occupational categories, and the demographic composition of the U.S. workforce have changed substantially since then. A model trained on 1994 data may not generalise well to contemporary donor populations.
- **Class imbalance:** Although the F-0.5 score partially addresses the 75/25 class split, no explicit resampling techniques (e.g., SMOTE, class weighting) were applied. Techniques such as `scale_pos_weight` in XGBoost could further improve recall of the minority class if CharityML's priorities shift.
- **Fairness and bias:** The dataset includes sensitive demographic features such as race, sex, and native-country. The model may encode historical socioeconomic biases present in 1994 U.S. census data. No fairness auditing (e.g., disparate impact analysis) was performed in this study.
- **Hyperparameter search scope:** The GridSearchCV covered 27 parameter combinations, a relatively small subset of XGBoost's full hyperparameter space. Bayesian optimisation or random search over a wider grid (subsample, `colsample_bytree`, `min_child_weight`) could yield further improvements.
- **Missing value imputation:** Simple median/mode imputation was applied to the Kaggle test set. More sophisticated imputation methods (e.g., KNNImputer, IterativeImputer) could reduce imputation bias, particularly for the 13–22 missing values per column observed.

10. Directions for Future Research

Several avenues of future research could extend and improve the current work:

- Deep learning approaches: Tabular deep learning architectures such as TabNet (Arik & Pfister, 2021) or FT-Transformer (Gorishniy et al., 2021) have recently achieved competitive or superior performance to gradient-boosted trees on select tabular datasets. A comparison of these architectures on the census income task would be informative.
- Ensemble stacking: Rather than selecting a single best model, a stacked ensemble combining the predictions of XGBoost, Random Forest, and AdaBoost as meta-features for a logistic regression meta-learner could potentially exceed the performance of any individual model (Wolpert, 1992).
- Fairness-aware classification: Future work should incorporate fairness constraints or post-processing calibration to audit and mitigate potential discriminatory outcomes across race, sex, and national origin — aligning the system with emerging AI ethics and regulatory requirements (Barocas et al., 2019).
- Updated datasets: Retraining on more recent census data (e.g., the American Community Survey, 2020–2024) would significantly improve the real-world applicability of the model to contemporary donor identification campaigns.
- Explainability: Integrating SHAP (SHapley Additive exPlanations) values (Lundberg & Lee, 2017) would provide individual-level prediction explanations, enabling CharityML to understand exactly why specific individuals are predicted as donors — increasing stakeholder trust and enabling targeted messaging.

11. References

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Appendix A: Full Model Comparison Summary

All metrics reported on the held-out test set (9,045 samples, 20% stratified split, random_state=0). F-score uses beta=0.5 (precision-weighted).

Model	Accuracy	F-0.5	Train Time	Notes
Naive Predictor	0.2478	0.2917	—	Always predicts >50K
Logistic Regression	0.8417	0.6826	~1s	Linear baseline
Random Forest	0.8423	0.6813	~8s	100 trees; overfits train
AdaBoost	0.8483	0.7029	~12s	Default n=50 stumps
XGBoost (untuned)	0.8694	0.7498	~15s	n=100, lr=0.1, depth=6
XGBoost (tuned) ★	0.8699	0.7507	~45s*	Best: lr=0.2, d=3, n=200

★ GridSearchCV: 27 combinations, 5-fold CV, F-0.5 scorer. *Includes grid search over $27 \times 5 = 135$ fits.

Green values indicate best performance in that column.